

PRE BOARD EXAM (2024-25)
CLASS- X
SUBJECT- ARTIFICIAL INTELLIGENCE (417)

Max. Time: 2 Hours

Max. Marks: 50

GENERAL INSTRUCTIONS:

1. Please read the instructions carefully.
2. This Question Paper consists of **21 questions** in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. **Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.**
5. All questions of a particular section must be attempted in the correct order.
6. **SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**
 - i. This section has 05 questions.
 - ii. Marks allotted are mentioned against each question/part.
 - iii. There is no negative marking.
 - iv. Do as per the instructions given.
7. **SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**
 - i. This section has 16 questions.
 - ii. A candidate has to do 10 questions.
 - iii. Do as per the instructions given.
 - iv. Marks allotted are mentioned against each question/part.

SECTION A: OBJECTIVE TYPE QUESTIONS

Q. 1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
i.	Sonika gets up at 6 am and goes to her hobby classes. Then she comes back home and finishes her homework before going to school. She does all work by herself. No one tells her to do so. Which self-management skill is clearly visible in the given statement? a. Self-Awareness b. Self-Motivation c. Self-Regulation d. Discipline	1
ii.	Which Operating System is used to control machineries like robots in complex animations and computer-controlled automated machines. a. Real Time b. Multi-user c. Single-user, multi-task operating system d. Distributed	1
iii.	Assertion (A): Entrepreneurship is the act of creating a new and innovative enterprise/ product or service. Reason(R): An entrepreneur must be creative, organized and risk taker. a. Both A and R are true and R is the correct explanation of A. b. Both A and R are true but R is not the correct explanation of A. c. A is true but R is false. d. A is false but R is true.	1

iv.	Stay current with best practices and read up on new information always. Prepare yourself for new changes, so that you can transition seamlessly. This skill is known as _____ a. Self-awareness b. Responsibility c. Time Management d. Adaptability	1
v.	You need to apply leave at work? Which method of communication will you use? (a) e-mail (b) Poster (c) Newsletter (d) Blog	1
vi.	Solution to these global problems like scarcity of clean water to drink, scarcity of pure air to breathe, scarcity of unadulterated food, rising issue of global warming is _____ a. Stop using these resources. b. Sustainability c. to make people aware d. None of the above	1

Q. 2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Assertion (A): Alex says to his students that we use many types of intelligence to perform well in our daily life tasks. Reason(R): John McCarthy proposed many types of intelligence such as linguistic intelligence, naturalistic intelligence, visual-spatial intelligence. a. Both A and R are true and R is the correct explanation of A. b. Both A and R are true but R is not the correct explanation of A. c. Both A and R are false. d. A is true but R is false.	1
ii.	Results based on discriminatory assumptions made while developing algorithm by the developer is called as _____. a. AI Ethics b. Test Data c. AI Bias d. Training Data	1
iii.	Assertion(A): Neural networks are the backbone of deep learning algorithms Reason(R): Neural networks use vast amounts of data a. Both A and R are correct b. Both A and R are not correct c. A is correct but R is not correct d. A is not correct but R is correct.	1
iv.	The makes the data understandable for humans as we can discover trends and patterns out of it. a) Random Data b) Graphical Representation c) Unstructured Data d) None of the above	1
v.	It not only gives us a clearer idea around the dataset, but also adds value to it by providing deeper and clearer analyses around it. a. Data Privacy b. Data Piracy c. Data Collection d. Data Science	1

vi.	A corpus contains 4 sentences in which the word 'such' was appearing frequently. Identify the term in which we can categorise the word 'such'. a. Stop word b. Rare word c. Frequent word d. Removable word	1
------------	---	----------

Q. 3	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	An air conditioner can be turned on and off remotely with the help of internet but still needs a human touch. This is an example of ____ a. Artificial Intelligence b. Internet of Things c. Machine Learning d. AI Biased	1
ii.	Which of the following uses dots to represent the relationship between two different numeric variables represented on the x and y axis? a. Histogram b. Bullet Graphs c. Scatter plot d. Tree Diagram	1
iii.	Which of the following is an application of data science? a. Genetics and Genomics b. Text summarization c. Face lock in smartphones d. Email filters	1
iv.	Which of the following tasks involves identifying and locating objects within an image? a. Image compression b. Feature extraction c. Object detection d. Image enhancement	1
v.	What is a chatbot called which uses simple FAQs without any intelligence? a. Smart Chatbot b. Script Chatbot c. AI Chatbot d. ML Chatbot	1
vi.	Which of the following parameters will be consider by recall, while evaluating a model's performance? i. False negative ii. True negative iii. False positive iv. True Positive Choose the correct option: a. only (i) b. (ii) and (iii) c. (iii) and (iv) d. (i) and (iv)	1
Q. 4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Alok learnt about AI terminologies but was not able to recollect the term that is a subtype of machine learning imitates how the human brain functions. It is modeled after the neurons found in the human brain and uses neural networks to tackle challenging real-world issues. a. Deep learning b. Machine learning	1

	c. Computer Vision d. Natural Language Processing	
ii.	Statement 1: The comparison between the results of Prediction and reality is called the Confusion matrix. Statement 2: There are only three cases/observations takes place in confusion matrix. a. Both Statement 1 and Statement 2 are correct b. Both Statement 1 and Statement 2 are incorrect c. Statement 1 is correct but Statement 2 is incorrect d. Statement 2 is correct but Statement 1 is incorrect	1
iii.	Sarthak made a face mask detector system for which he had collected the dataset and used all the dataset to train the model. Then, he used the same data to evaluate the model which resulted in the correct answer all the time but was not able to perform with unknown dataset. Name the concept. a. Object detection b. Sentiment analysis c. Overfitting d. Data visualization	1
iv.	A grayscale image represents intensity values ranging from 0 to _____. a. 256 b. 25.6 c. 254 d. 25.5	1
v.	Companies can use _____ in a lot of ways such as to find out the emotions of their target audience, to understand product reviews, to gauge their brand. a. Sentiment Analysis b. structure Analysis c. Deep Learning d. Emails	1
vi.	Which of the following statements is not true about overfitting models? a. This model learns the pattern and noise in the data to such extent that it harms the performance of the model on the new dataset b. Training result is very good and the test result is poor c. It interprets noise as patterns in the data d. The training accuracy and test accuracy both are low	

Q. 5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Rohan, a student of class XII developed a chatbot that clarifies the doubts of Geography students. He trained the software with lots of data sets catering to all difficulty levels. If any student would type or ask questions related to Geography, the software would give an instant reply. Identify the domain of AI in the given scenario. a. Computer Vision b. Data Science c. Natural Language Processing d. None of these	1
ii.	The percentage of true positive cases versus all the cases where the prediction is true is defined as: a. Precision b. Accuracy c. F1 Score d. None of these	1

iii.	What does the pixel value represent in a grayscale image? a. The color b. The intensity c. The contrast d. The brightness	1
iv.	Which of the following statement is incorrect? a. Email Filtering is an application of NLP b. Price comparison websites are applications of Data Science c. Virtual Assistants are not applications of NLP d. Self-driving cars are applications of Computer Vision.	1
v.	TF-IDF in NLP is defined as: a. Term Frequency and Definite Frequency b. Term Frequency and Indefinite Frequency c. Term Frequency and Inverse Document Frequency d. Term Frequency and Integrated Document Frequency	1
vi.	Seema is learning the conditions that make up the confusion matrix. She came across a scenario in which the machine that was supposed to predict a bird was always predicting not a bird. What is this condition called? a. False Positive b. True Positive c. False Negative d. True Negative	1

SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks)

Answer each question in 20 – 30 words.

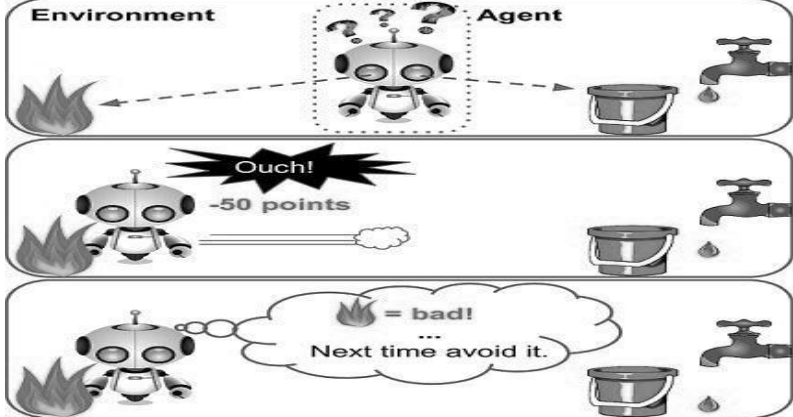
Q. 6	Enumerate any two ways to overcome barriers to effective communication.	2
Q. 7	In SMART goals, what does 'S' stand for? Explain.	2
Q. 8	Aisha is searching some contents related to sports in search engine. Help her how to prevent her data while using the internet?	2
Q. 9	How does an entrepreneur promotes economic prosperity of a country?	2
Q. 10	"With increasing population and income, the consumption of goods is increasing day by day." – Comment?	2

Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)

Q. 11	How do you understand whether a machine/application is AI based or not? Explain with an example.	2
Q. 12	Gaurav want to make a model which will organize dataset comprising of 100 images of apples and 100 images of bananas. It compare the image with the trained data and according to the labels of trained images, it will identify the test image as an apple and banana. Which learning model should he use and why?	2
Q. 13	What are the different types of sources of data from where we can collect reliable and authentic datasets?	2
Q. 14	Explain how pixel density affects the visual quality of an image.	2
Q. 15	Identify the stop words in the given sentence: Pollution is the introduction of contaminants into the natural environment that cause adverse change. The three types of pollution are air pollution, water pollution and land pollution. Mail details pollu@gmail.com	2

Q. 16	Draw the confusion matrix for the following data • the number of true positive = 10 • the number of true negative = 5 • the number of false positive = 15 • the number of false negative = 3	2
--------------	---	----------

Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)

Q. 17	Differentiate between what is AI and what is not AI with the help of an example?	4													
Q. 18	What is 4Ws Problem Canvas? Explain Problem Statement Template.	4													
Q. 19	Identify and explain the types of the learning-based approaches in the figures given below. Environment Agent  1 Observe 2 Select action using policy 3 Action! 4 Get reward or penalty 5 Update policy (learning step) 6 Iterate until an optimal policy is found	4													
Q. 20	What is the difference between human language and computer language.	4													
Q. 21	An IT company situated in Bombay developed an AI model which predicts the purchasing of electronic gadgets. During testing, the AI model came up with the following predictions. Based on the given predictions, calculate the following:- <table><tr><th colspan="2" rowspan="2">Confusion Matrix</th><th colspan="2">Reality</th></tr><tr><th>Yes</th><th>No</th></tr><tr><th rowspan="2">Prediction</th><th>Yes</th><td>60</td><td>25</td></tr><tr><th>No</th><td>5</td><td>10</td></tr></table> i. How many total tests have been performed in the above scenario? ii. Calculate precision, recall and F1 score.	Confusion Matrix		Reality		Yes	No	Prediction	Yes	60	25	No	5	10	4
Confusion Matrix				Reality											
		Yes	No												
Prediction	Yes	60	25												
	No	5	10												